

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2023-24

Botany**Paper: BOB-201 — Microbiology, Plant Pathology, Cytology and Genetics**

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all, selecting two each from Sections A and B and Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following questions in not more than 50 words each: $[10 \times 1 = 10]$

- (a) Give one example of broad spectrum antibiotics.
- (b) Write the role of infection threads in *Ustilago tritici*.
- (c) Write about the inoculum of powdery mildew of pea.
- (d) What is Hfr?
- (e) Write one major difference between the flagellum of Gram-positive and Gram-negative bacteria.
- (f) Name the stage of cell division during which crossing-over occurs.
- (g) Write the role of cytoskeleton in cell division.
- (h) What is the purpose of a test cross?
- (i) Define recon.
- (j) What do you mean by duplicate gene?

Section-A**Question 2.** Write short notes on any two of the following: $[2 \times 7.5 = 15]$

- (a) Molecular criteria for bacterial classification
- (b) Bacterial cell wall
- (c) Disease management practices for Wilt of Pigeon pea

Question 3. Write short notes on any two of the following: $[2 \times 7.5 = 15]$

- (a) Disease cycle of Late Blight of potato
- (b) Role of microbes in production of alcoholic beverages
- (c) Symbiotic nitrogen fixation

Question 4. What is recombination? How are the genes transferred from one bacterial cell to another by transduction? Describe in detail with illustration. [15]

Question 5. Discuss the structure and life cycle of Lambda bacteriophage. [15]

Section-B

Question 6. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Synaptonemal complex
- (b) Nuclear pore complex
- (c) Inhibitory genes

Question 7. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Nucleosome
- (b) Neo-Mendelism
- (c) Law of segregation

Question 8. Write a detailed account of sex determination in plants. [15]

Question 9. What is linkage? Discuss in detail with suitable examples. [15]

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